

MAR 6930 · Marketing Data Visualization

Introduction to data visualization

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Some materials in these slides are from Knaflic, C. N. (2015), *Storytelling with Data*.

Goals

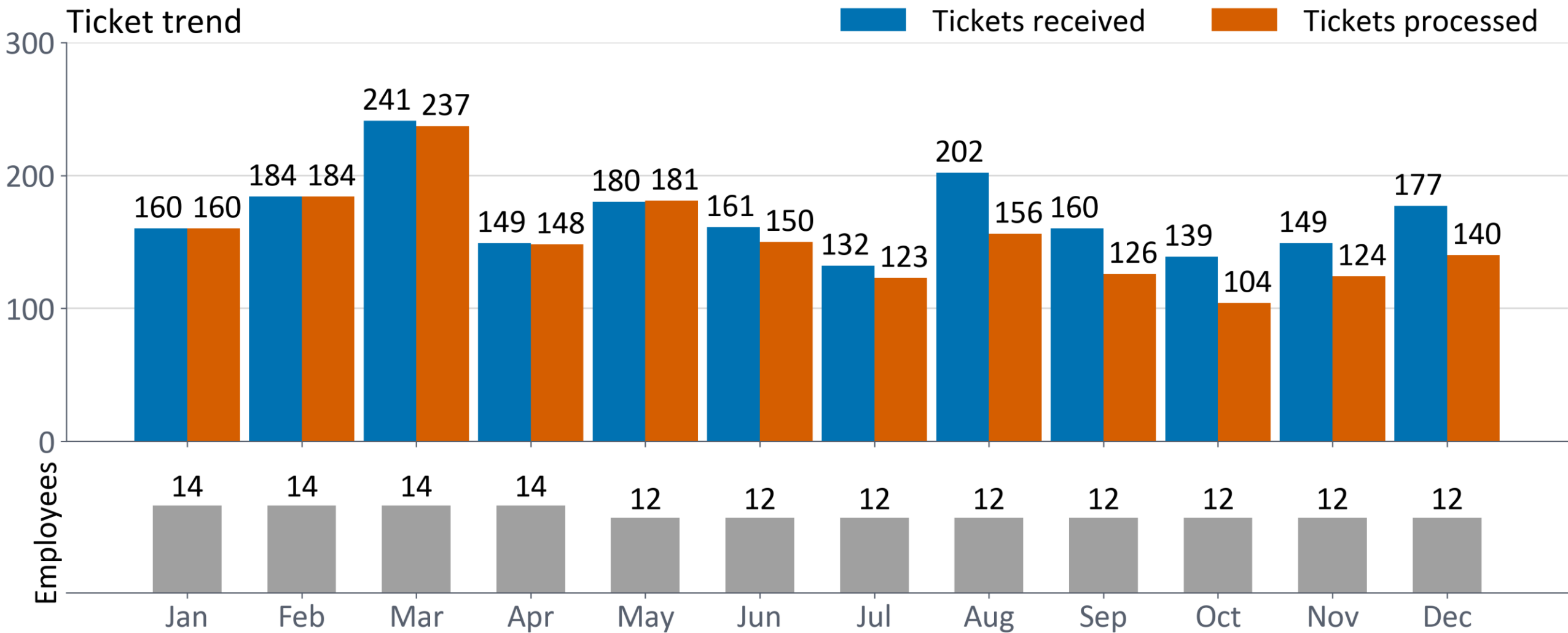
1. The objective of visualization is to get the point across.
2. Good visuals draw on psychology, data thinking and business understanding.
3. Go from a business question to a chart.
4. Set up environment and run your first notebook in VS Code.

What can you learn from this table?

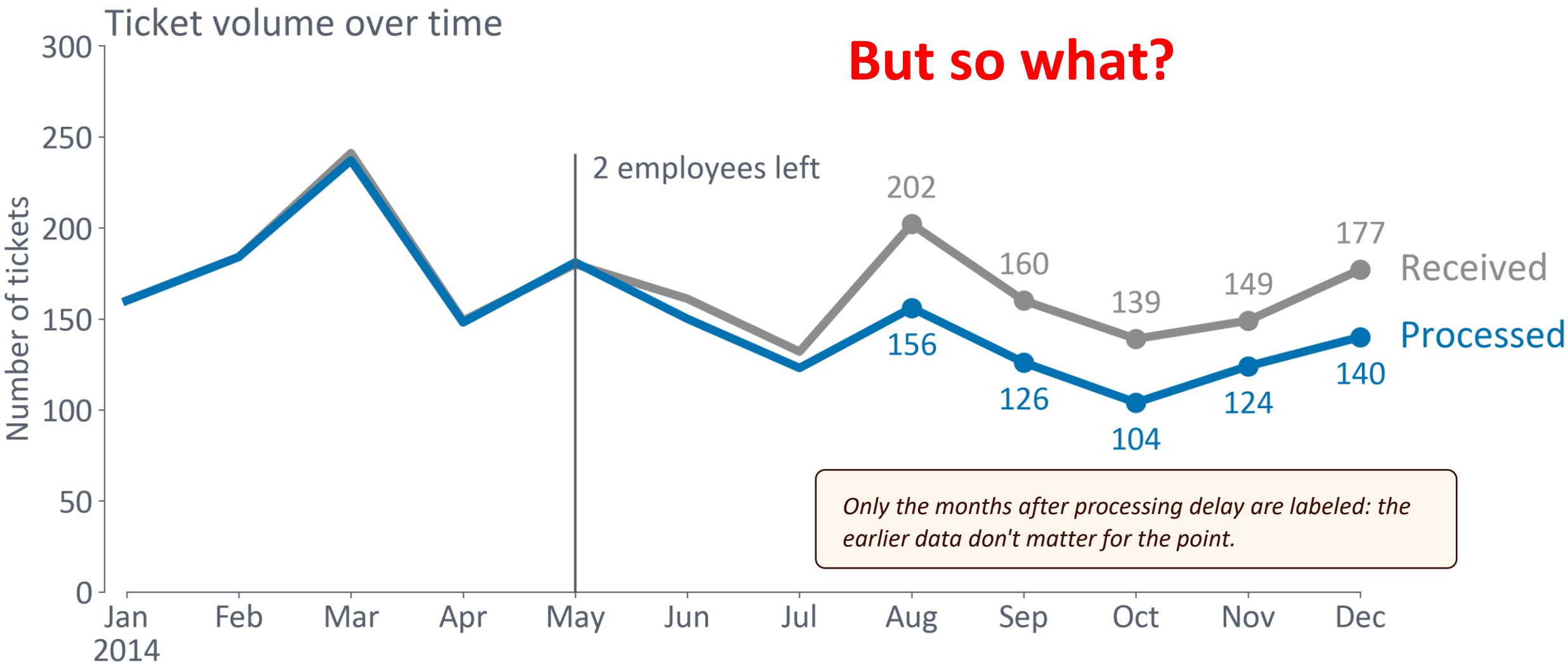
Month	Tickets received	Tickets processed	Employees
Jan	160	160	14
Feb	184	184	14
Mar	241	237	14
Apr	149	148	14
May	180	181	12
Jun	161	150	12
Jul	132	123	12
Aug	202	156	12
Sep	160	126	12
Oct	139	104	12
Nov	149	124	12
Dec	177	140	12

What can you learn from this table?

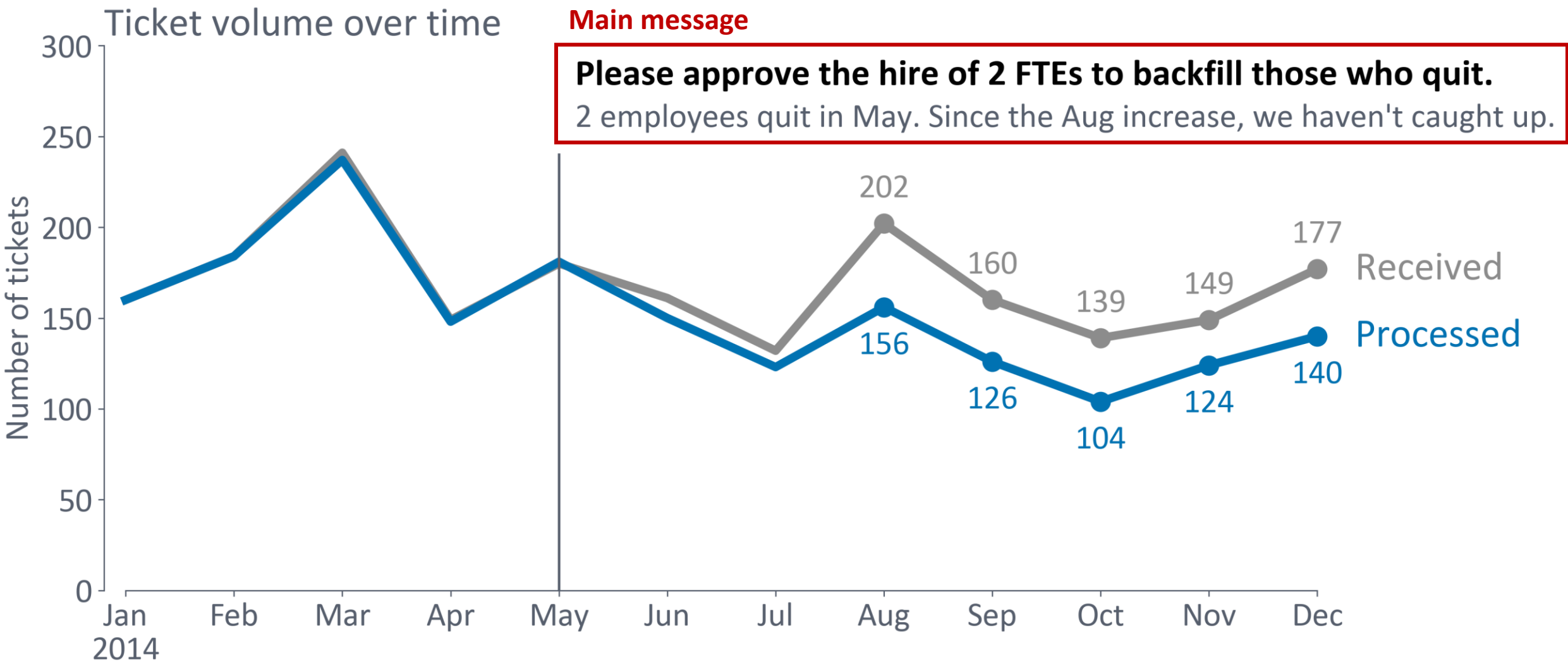
Visualization: what does the plot tell you?



Better visualization: highlight what matters



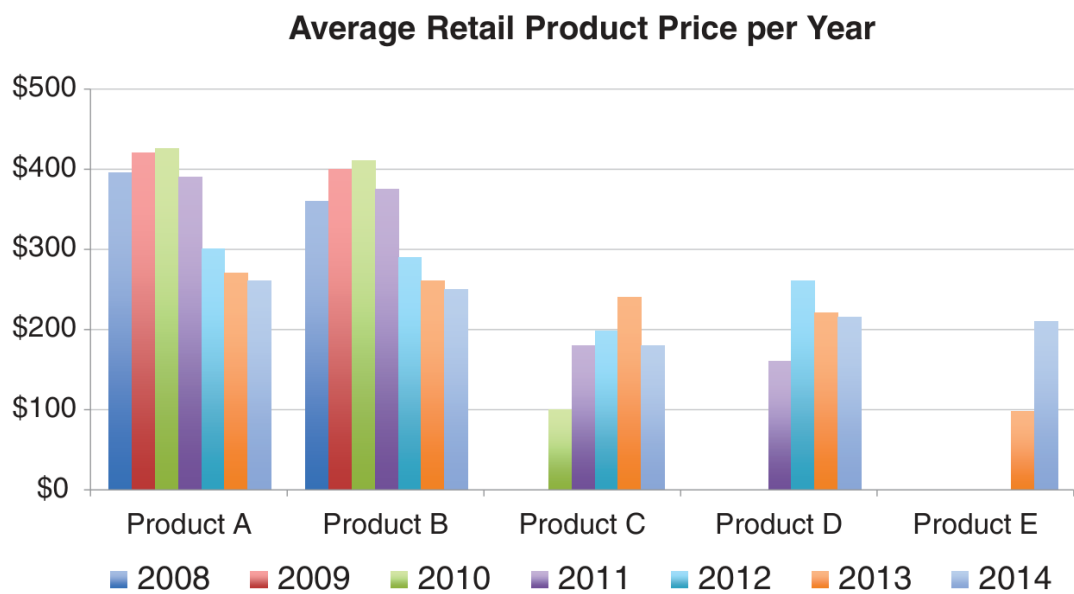
Even better visualization: deliver a business suggestion



Data visualization tells a story

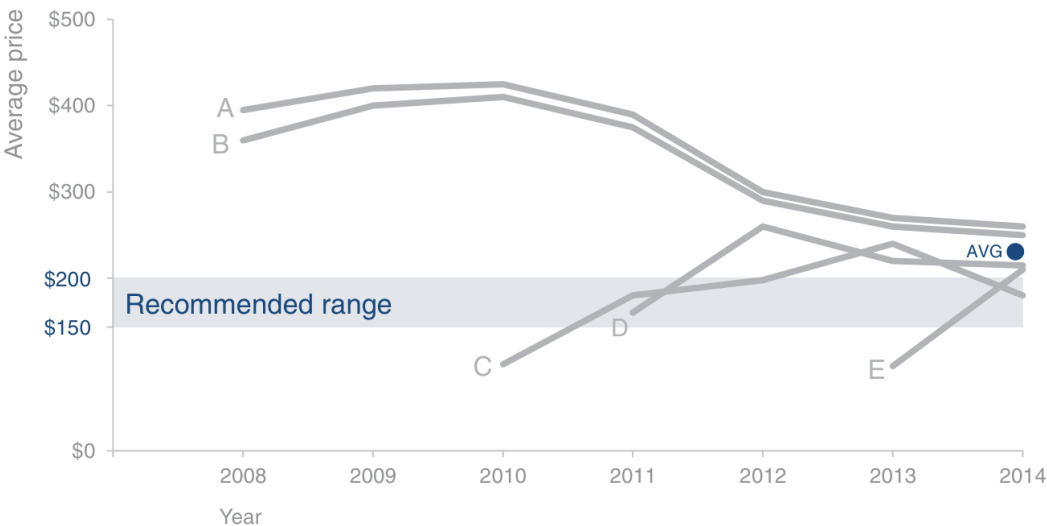
- The goal is to tell a story: convey a point or make a suggestion to your audience.
- Let them get it quickly and correctly, and remember it.
- Data visualization is not limited to describing facts.
- Your job: generate insights from your data and find the right way to visualize them.

Discussion: what is wrong with the first chart, and how does the second fix it?

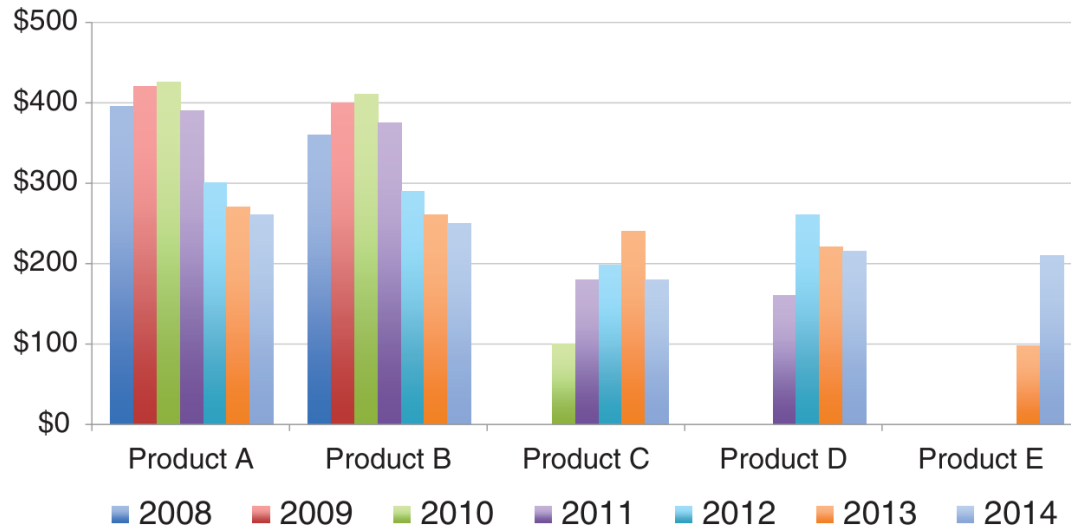


To be competitive, we recommend introducing our product *below the \$223 average price point in the \$150–\$200 range*

Retail price over time by product

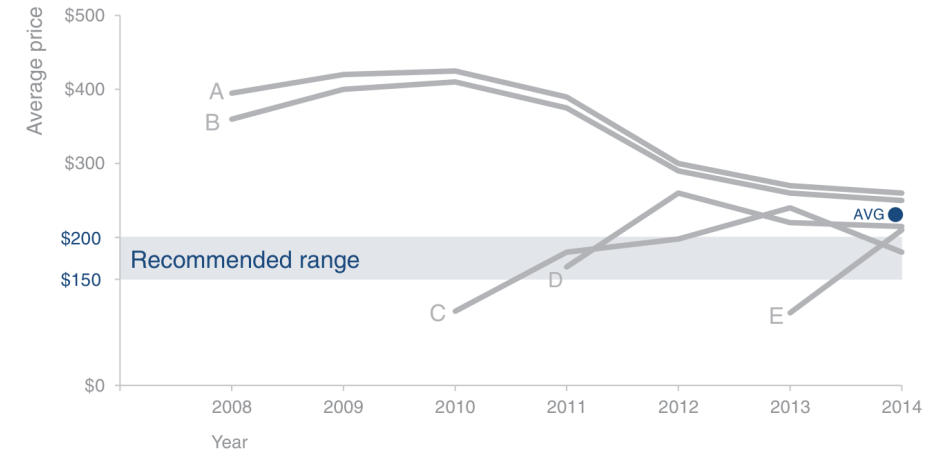


Average Retail Product Price per Year



To be competitive, we recommend introducing our product *below the \$223 average price point in the \$150–\$200 range*

Retail price over time by product



Leads with a decision

Recommends a \$150–\$200 launch price, not just past prices.

Uses color with purpose

Gray for context; blue only for the recommendation.

Fits the chart to the data

Lines show change over time better than grouped bars.

Cuts clutter

No border, gridlines, heavy formatting or oversized title.

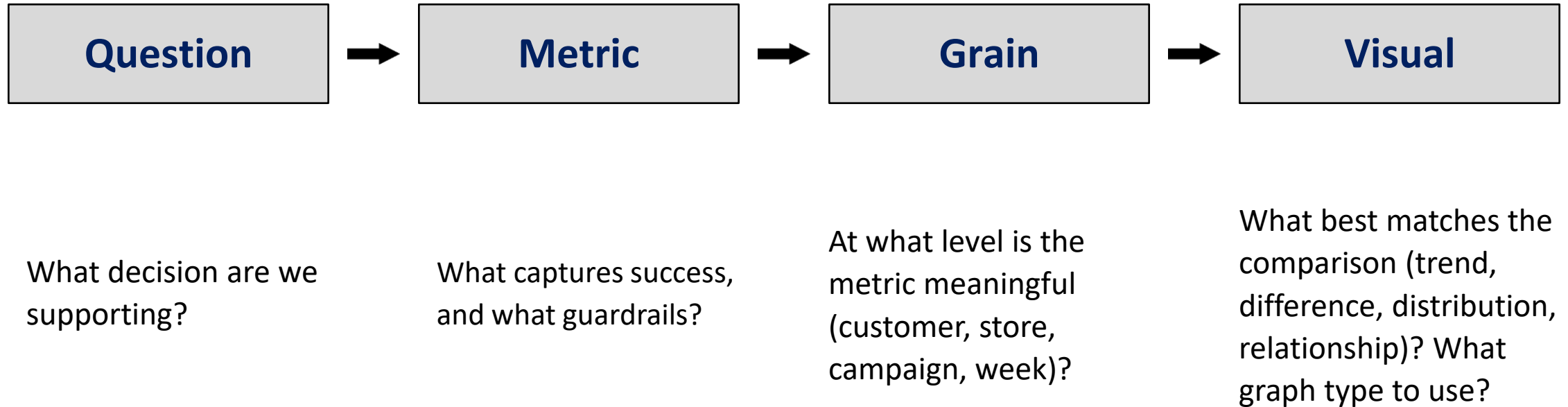
Gets the conclusion right

Not every price fell after 2010; the lines show the real patterns.

Adds reference points

The shaded range and \$223 average tie the history to the decision.

Repeatable framework: question, metric, grain, visual



We practice this framework every session.

Your output is a recommendation supported by a plot, not "a plot".

Rule of thumb: if you can't state the decision in one sentence, you're not ready to plot.

Example decision: which channel should we advertise in?

Decision: which channel should we advertise in: Paid Search, Paid Social, Email or Organic search?

- **Metrics (per channel):** cost per acquired customer (\$), spending per customer, profit margin per customer
- **Over time:** how each metric changes in the weeks after launch (campaign effect decay)
- **Grain:** channel-week
- **Visuals:** bar for cross-sectional comparison (channels at one point in time); line for variation over time

Example insight will be like: the cheapest channel to acquire customers today may not be the most profitable once its effect decays.

Grain: the level of detail you look at changes the answer

Definition: grain (granularity) is the level of detail at which you look at the data, in time in time and in place.

Time: day vs week

Daily numbers show noise and day-to-day swings; weekly numbers show the trend.

Place: store vs market

A market total can hide which stores are winning and losing.

Your turn: our brand's revenue is 112 daily numbers or 16 weekly totals over the same 16 weeks. For a weekly marketing meeting, which do you want?

Exercise: match each question to the chart that answers it

Data: daily revenue of 10 stores for one year (52 weeks). Deliverable: your three matches and one reason each.

What point to make

(a) How does the holiday season lift sales compared with the normal season?

→ (2)

(b) How do weekend sales compare with weekday sales?

→ (3)

(c) Which are the top 3 stores by revenue?

→ (1)

What graph to make

(1) Total revenue per store for the year: sorted bars, top 3 highlighted

(2) Monthly sales of all stores combined: line chart, holiday months marked

(3) Average daily revenue per store, weekend vs weekday: two bars

Tools for this course

- **Visualization programming tool:** Python (VS Code).
- **AI coding assistant:** GitHub Copilot, free education version (registration steps in the setup section).
- You can also use ChatGPT (Codex) or Claude Code if you already have a \$20/month subscription to either.
- We will discuss agentic AI for automating visualization tasks, if time allows (optional).
 - Works best with Codex or Claude Code (paid).
 - Tasks run in their desktop apps; VS Code is used to check the code.
 - A paid AI subscription is not needed for the course.
- **Free AI chatbot:** UF NaviGator.

Understand what the code does; you don't need to memorize it

What you don't need

- You don't have to memorize how to code everything from scratch.
- AI and web search can handle most technical details.

What you do need

- You do need to understand what the code is doing.
- You do need to know what graph you want AI to draw
- Link the code to the purpose of the visualization task and to the concepts from class.

Power BI and Tableau: not used here, but the concepts carry over

- Other popular visualization software: Power BI and Tableau.
- They are free for basic features, but full functions (e.g., AI) may require payment.
- We don't use them in this class, but the concepts we learn carry over.
 - With AI, learning to operate them is easy these days.

Use AI tools, but understand and own the output

- You are encouraged to use AI tools (chat, coding assistants).
- We will learn how to use coding agents for visualization throughout the course.
- It is important to understand the AI output: use it to aid your understanding of the course material, not to replace it.
- **You are responsible for the final output.**

When nobody checks the AI output:

ECONOMY

State Farm defense lawyers admit AI generated fake cases in LA lawsuit

 BY LEVI SUMAGAYSAY
AUGUST 18, 2026

Republish   

A.I. ‘Hallucinations’ Created Errors in Court Filing, Top Law Firm Says

Sullivan & Cromwell apologized for submitting a court document that had fake citations created by artificial intelligence.

How you will be evaluated

Component	Points
Two in-class exams	60 (30 each)
Project presentation (all group members present, last week)	20
Class participation (quality, not quantity)	20

- We go through Python code so you understand what happens behind a visualization.
- Sample Python code will be provided.
- AI can generate code, but you are responsible for its output.
- Note: Many company interviews do not allow AI tools.

Apply for GitHub Education

Unless you already have Copilot, use this page: [Apply to GitHub Education as a student](#)

 GitHub Docs

Version: Free, Pro, & Team

Search or ask Copilot

← Home

GitHub Education

Quickstart for educators

About GitHub Education

For students

About

Apply to GitHub Education

Troubleshooting access

About Community Exchange

Terms for Student Benefits

For teachers

At your institution

GitHub Community Exchange

GitHub Classroom

Guides

GitHub Education / About GitHub Education / For students /

Apply to GitHub Education as a student

Unlock tools and resources for your educational journey.

Do I qualify for GitHub Education?

You qualify for GitHub Education if you:

- Are enrolled in a degree- or diploma-granting program, such as a high school, college, university, or homeschool.
- Have a verifiable school-issued email or documents proving your student status.
- Own a [GitHub personal account](#).
- Are at least 13 years old.

What documents can I use to prove I'm a student?

- A picture of your school ID with current enrollment date
- Your class schedule
- Your transcript
- An affiliation or enrollment verification letter

Applying to GitHub Education

- 1 Navigate to your [Education benefits](#) settings.
- 2 Under "GitHub Education", click **Start an application**.
- 3 Complete the form, then click **Submit application**.

Once your application is approved, navigate to your [GitHub Education portal](#) to use your GitHub Education benefits, such as **free access** to professional developer tools.

Further reading

- [Solving problems with your GitHub Education access](#)

In this article

Do I qualify for GitHub Education?

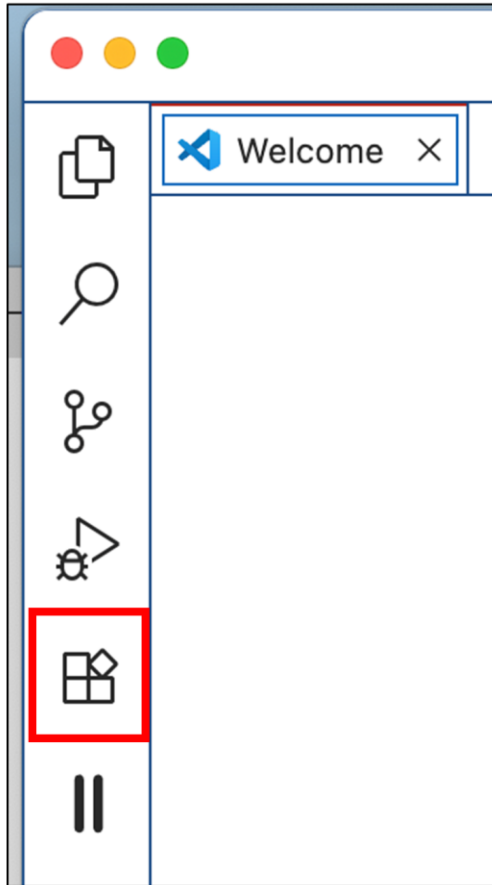
What documents can I use to prove I'm a student?

Applying to GitHub Education

Further reading

Use your **ufl.edu** email address to verify your eligibility.

Install the VS Code extensions

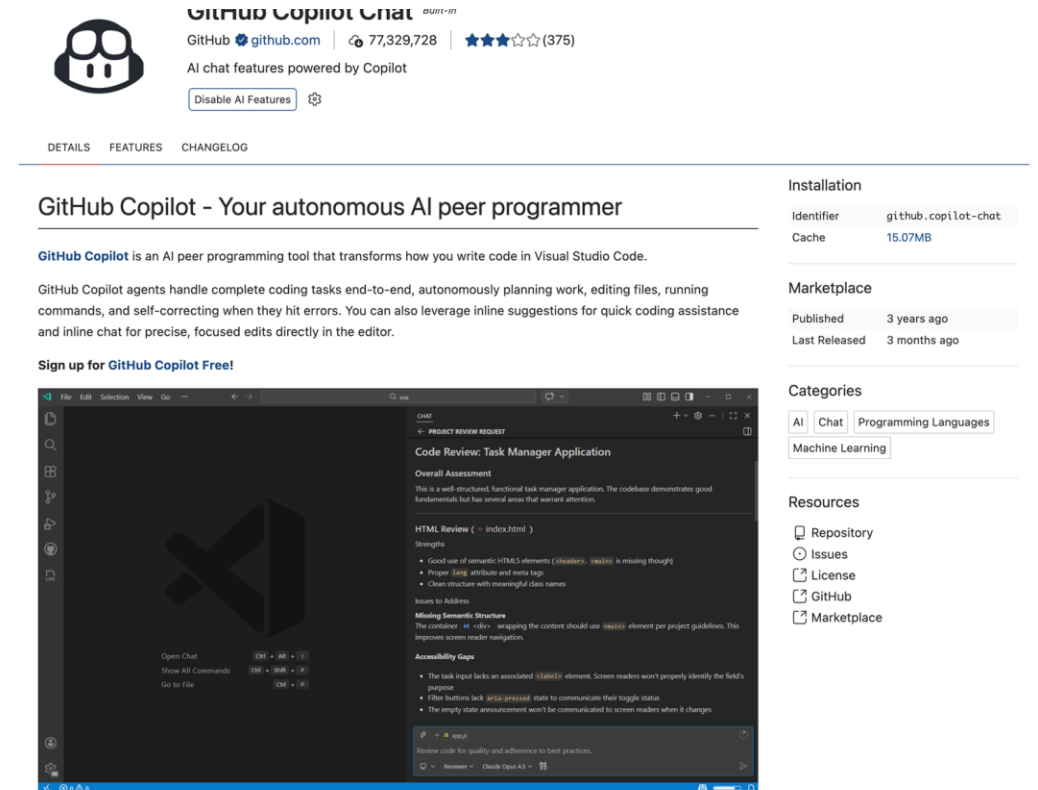


Extensions to install:

- GitHub Copilot Chat (now built-in)
- Python (Microsoft)
- Jupyter (notebook support)
- Jupyter Notebook Renders

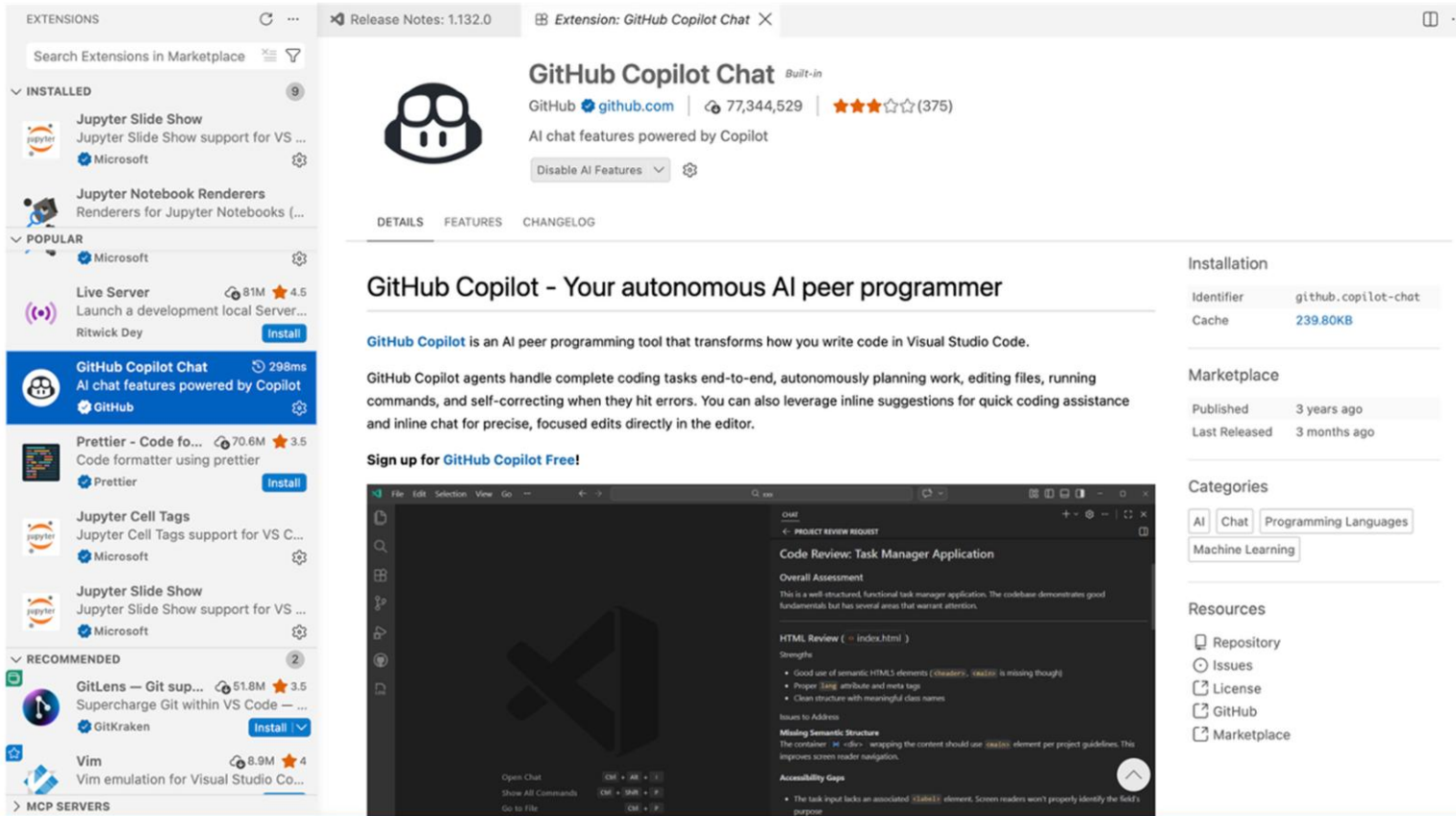
Optional Extensions to consider:

- Claude Code for VS Code
- Codex (OpenAI) for VS Code



Set up GitHub Copilot in VS Code

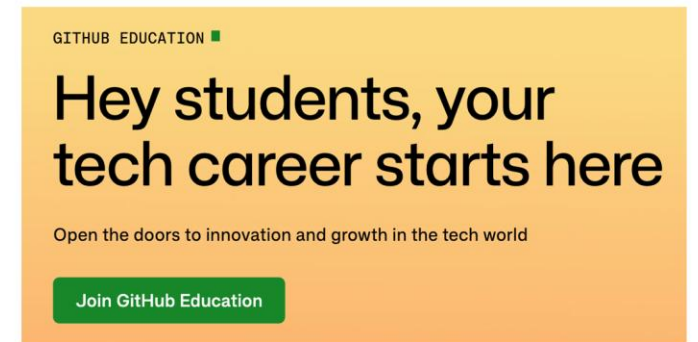
Setup guide: code.visualstudio.com/docs/setup/copilot



The screenshot shows the VS Code Extensions Marketplace interface. On the left, the 'EXTENSIONS' sidebar is visible with a search bar and a list of installed and popular extensions. The 'GitHub Copilot Chat' extension is highlighted in blue. The main panel displays the extension's details, including its name, version (1.132.0), and a brief description: 'AI chat features powered by Copilot'. Below this, there's a section titled 'GitHub Copilot - Your autonomous AI peer programmer' with a detailed description of its capabilities. The right sidebar shows the 'Installation' section with the identifier 'github.copilot-chat' and a cache size of '239.80KB'. Below that, the 'Marketplace' section shows the extension was published '3 years ago' and last released '3 months ago'. The 'Categories' section lists 'AI', 'Chat', and 'Programming Languages'. The 'Resources' section includes links to the 'Repository', 'Issues', 'License', 'GitHub', and 'Marketplace'.

Free Account for Students

<https://github.com/education/students>



The banner features the GitHub Education logo at the top left. The main text reads 'Hey students, your tech career starts here' in a large, bold, black font. Below this, a smaller line of text says 'Open the doors to innovation and growth in the tech world'. At the bottom right, there is a green button with the text 'Join GitHub Education'.

Open Copilot for the first time

First-Time Activation Steps

If you do not see the icon or shortcuts working, ensure the extension is active and authenticated:

Install the Extension: Open the Extensions tab (Ctrl + Shift + X), search for GitHub Copilot, and click **Install**.

- **Sign In:** Click the **Accounts icon** (bottom-left profile silhouette) or look for the popup prompt to **Sign in with GitHub**.
- **Authorize:** Authorize the connection in your browser when prompted to sync your active Copilot subscription or Copilot Free trial.
- **Verify Status:** Check the small Copilot icon in the bottom-right Status Bar to ensure it is connected and active.



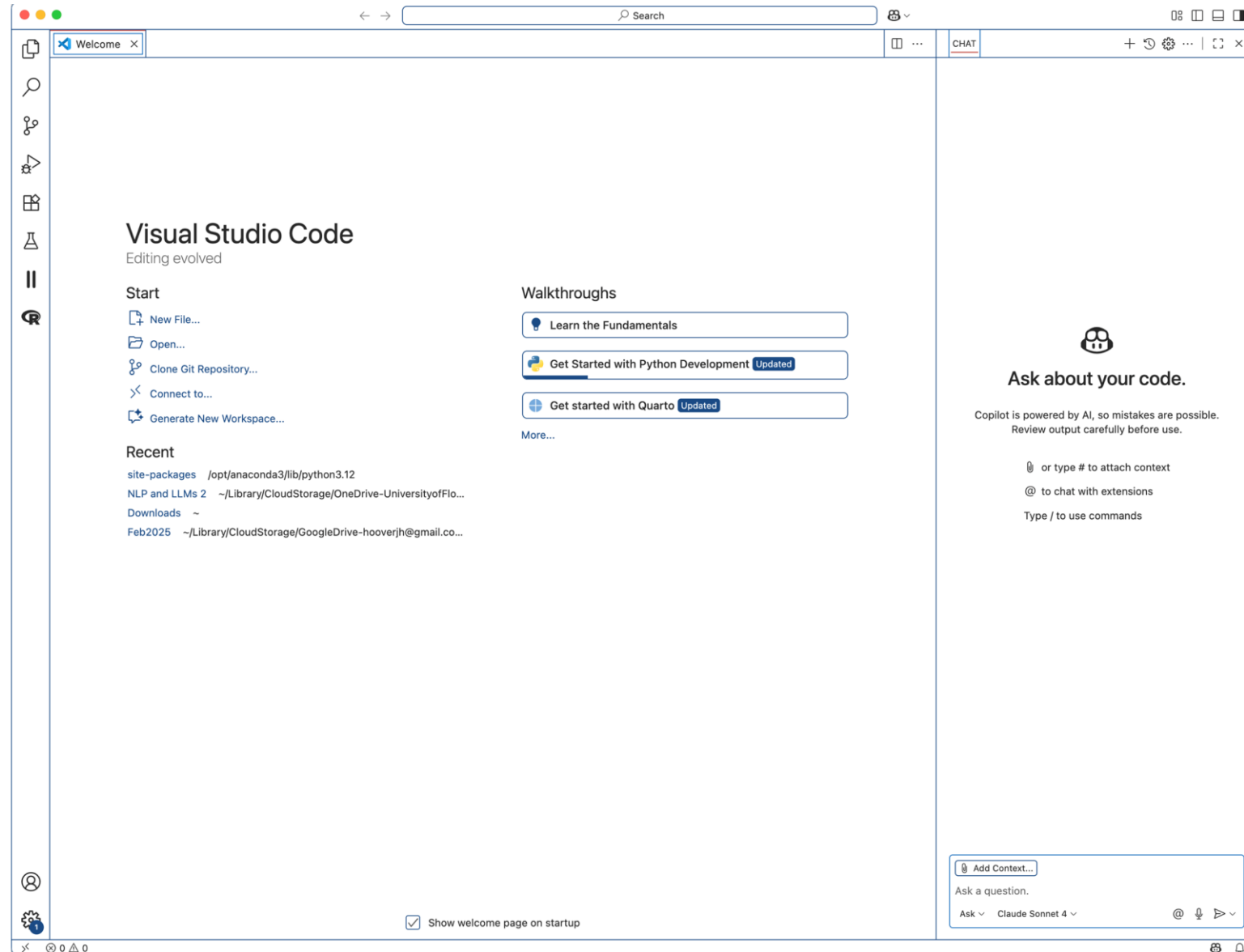
To open and use GitHub Copilot in Visual Studio Code, press **Ctrl + Alt + I (Windows/Linux) or Cmd + Ctrl + I (Mac)** to immediately launch the Copilot Chat panel.

If Copilot is not appearing or you are setting it up for the first time, follow the structural steps below to open it.

Quick Methods to Open Copilot

- **Keyboard Shortcut:** Press Ctrl + Alt + I (Cmd + Ctrl + I on Mac) for the Chat panel.
- **Activity Bar:** Click the **Copilot chat bubble icon** in the left-hand sidebar.
- **Command Palette:** Press Ctrl + Shift + P (Cmd + Shift + P on Mac), type Copilot, and select your desired action.
- **Inline Chat:** Press Ctrl + I (Cmd + I on Mac) directly inside any code file to open a prompt right at your cursor.

VS Code with Copilot Chat open



Activate Copilot Student

Access GitHub Copilot for free as a student

Get free access to Copilot's premium features as a verified student.

About Copilot Student access [↗](#)

Verified students on GitHub Education get **free access to Copilot's features**. See [About individual GitHub Copilot plans and benefits](#).

To become a verified student, see [Apply to GitHub Education as a student](#).

GitHub reevaluates your eligibility every month.

Access Copilot Student [↗](#)

- 1 Navigate to <https://github.com/settings/education/benefits>.

If you are not yet a verified student, click **Start an application** and complete the verification process.

- 2 Under "Free GitHub developer resources for students and teachers", click **Learn more**.

- 3 Follow the prompts to activate Copilot Student, configuring the Copilot use policies to meet your needs.

Troubleshooting [↗](#)

Account shows Copilot Free, a trial, or a paid checkout page after your GitHub Education verification is approved [↗](#)

Approval and Copilot activation are separate steps. The student benefit can take several days to finish applying after verification.

While you wait, you can try activating your included plan using one of the following steps:

- Navigate to your [Copilot settings](#) and follow the prompts.
- Go directly to https://github.com/github-copilot/free_signup.

If only paid options appear, do not complete a purchase. Wait a few days and try again. If the problem continues after several days, contact [GitHub Support](#).

Cautions:

- You need to understand what the code is doing
- You must evaluate the outputs of the code
- For your job interview, you might not have access to the tool
- For quizzes and tests at UF, you may not have access to the tool

Your first notebook: a brand's weekly marketing meeting

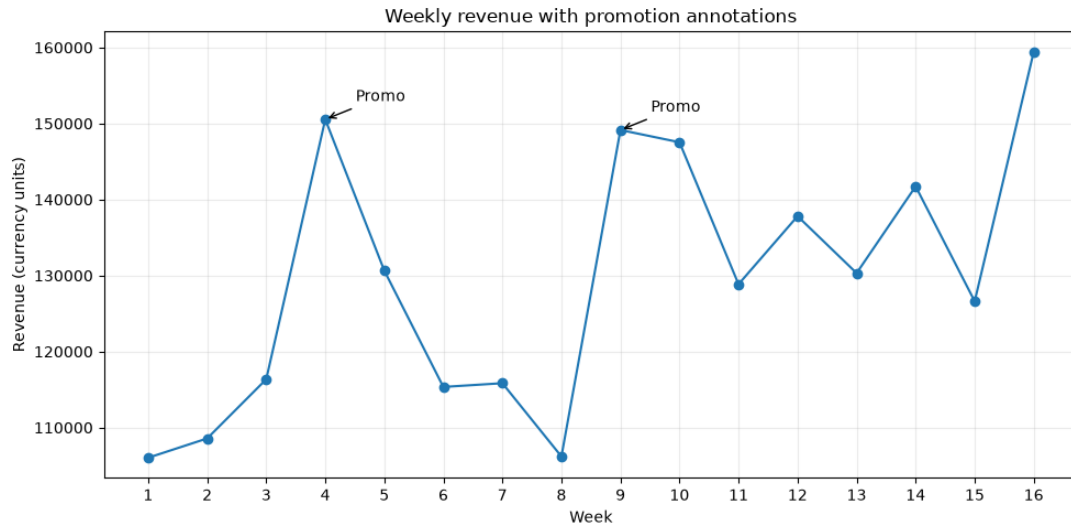
The manager's question

"Over the past 16 weeks, has revenue been growing, flat, or declining? Are we keeping our margin when revenue rises?"

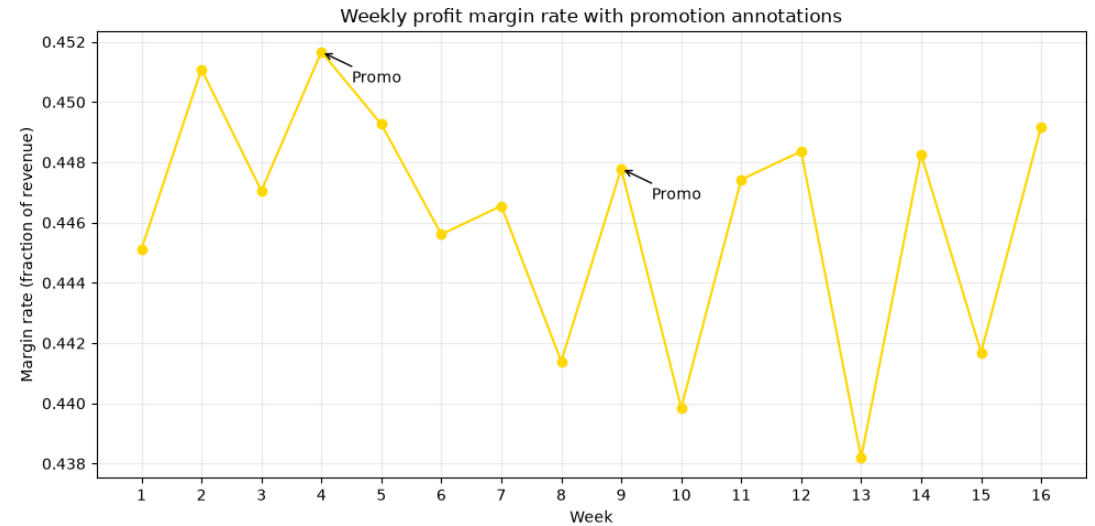
- **The data:** a brand's marketing performance dataset, with daily revenue for four channels (Paid Search, Paid Social, Email, Organic search), Sep 1 – Dec 21, 2025.
- **To run it:** put lecture1.ipynb and lecture1.csv in the same folder, open the notebook in VS Code, select the Python kernel, and run the setup cell (the one that imports pandas).

Check: the setup cell prints "Setup works. You are ready to load the data."

Revenue grew while the profit margin rate held steady



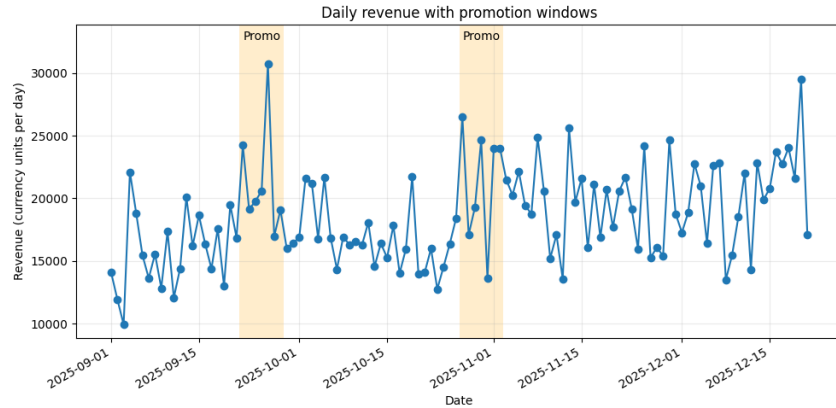
Weekly revenue rose over 16 weeks, with peaks in the promotion weeks.



The weekly profit margin rate stayed nearly constant, between 43.8% and 45.2%, including the promotion weeks.

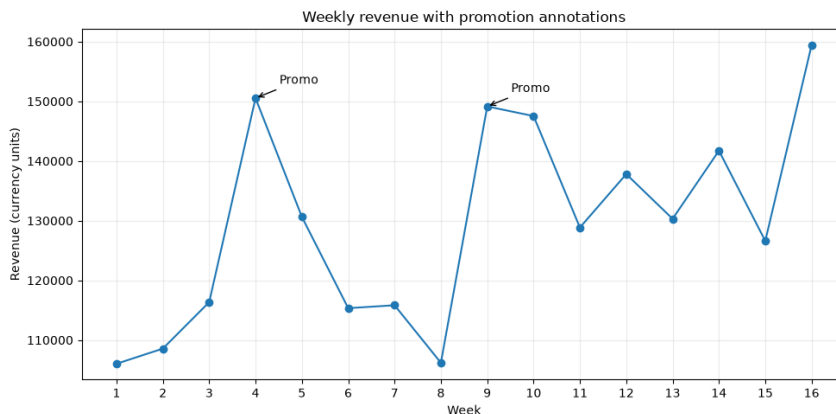
Daily points are the wrong grain for a weekly question

Daily (wrong grain here)



The daily chart includes day-to-day ups and downs that obscure the weekly promotion effects.

Weekly



Check: what does one point on my chart represent, and does it match the decision?

AI coding assistant for data visualization

- AI tools (e.g., GitHub Copilot) help write the code for visualization, but you still decide what it analyzes and how it shows it.
- As we have shown in previous examples, without proper guidance, AI can generate "visualization slops" failing your communication goals.

An experiment: compare the same AI tool's performance with different human guidance

- Each group gets its own folder (L1-A or L1-B) with only lecture1.csv.
- Start a new Copilot chat with the same model and mode.
- Paste the prompt once, then allow up to 10 minutes with no hints, revisions or regenerating.
- **Approve for edits:** when Copilot asks to create files or run code, approve it, or it stops and waits.

Two groups, two prompts for AI

Group A: one sentence

Using lecture1.csv, create a Python visualization to assess the brand's revenue and margin performance during the promotions on September 22–28 and October 27–November 2, 2025.

Group B: detailed

Create a short Python notebook using pandas and Matplotlib to assess promotion performance for a brand's weekly marketing meeting. Use lecture1.csv. Promotions ran September 22–28 and October 27–November 2, 2025, inclusive.

Each row is one channel on one day. Revenue and margin are currency amounts; promo is 1 on promotion days and repeats across channels.

Combine all channels into one row per week. Sum revenue and margin, then calculate margin rate as total margin divided by total revenue. Draw separate line charts of weekly revenue and weekly margin rate, with clear titles and units. Mark the two promotion periods and show all 16 weeks so we can compare them with surrounding weeks. Count promotion dates once, not once per channel.

Use a relative CSV path and leave the data unchanged. Check missing values, duplicate date-channel records, and whether aggregation preserves total revenue. Run the notebook and report errors. Finish with two observations about revenue and margin during promotions versus surrounding weeks.

Distinguish observed changes from causal effects: this comparison alone cannot prove that promotions caused a change.

Summary

1. Good visualization gets the point across: one message, ending in what the audience should do (ticket and price examples).
2. Success draws on psychology (compare lengths, not areas), data-driven thinking and business thinking.
3. From a business question to a chart: question → metric → grain → visual (matching exercise; daily vs weekly).
4. Your first notebook runs in VS Code; AI writes code, but you own the output.